channels, across several platforms and regions like North America, Europe, Japan, and other demographies for a decade. With the outbreak of the pandemic and different socio-political situations in various countries, the company, after going through various calculations and ledger books, thought of going through the heuristic sales figures to understand the impact (of the pandemic) in order to strategise marketing, capital management, inventory, sales channels, and other areas for the coming financial quarters. The data set as it appears in Figure 1 is continuous in nature.

The following are the different fields of interest that were captured and consolidated by the marketing analysts:

- Rank - Ranking of overall sales
- Name - The game's name
- Platform - Platform of the game's release (i.e., PC, PS4, etc)
- Year - Year of the game's release
- Genre - Genre of the game
- Publisher - Publisher of the game
- NA_Sales - Sales in North America (in million)
- EU_Sales - Sales in Europe (in million)
- JP_Sales - Sales in Japan (in million)
- Other_Sales - Sales in the rest of the world (in million)
- Global_Sales - Total worldwide sales

As the data is of continuous nature, we can take the route of supervised learning and select a regression analysis to predict the sales volume in the next year in different regions. This template pipeline could subsequently be executed on the model identified to achieve the expected accuracy. There could be several statistical models relevant to regression analysis, which might be actually compared to finalise the most desirable model, e.g., decision tree, SVM, Naive Bayes classifiers, etc.

As we go through the data sample and the ways for prediction, it seems we have a continuous data set where the supervised learning technique could be relevant. The following are the most frequently used models that are part of the supervised learning technique. For the sake of brevity, we will use the regression technique to derive the outcome. For the data sample referred here, we will consider linear regression, comparing the various independent variables against a dependent variable to predict a unit of range. The techniques itself, of course, can vary differentially between algorithms based on the continuous or discrete nature of the data sets or categorical data.

- Regression
- Logistic regression
- Classification

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
## Importing the dataset
dataset = pd.read_csv('../input/videogamesales/vgsales.csv')
```

Figure 2a: Google Colab sample Python notebook code for regression analysis

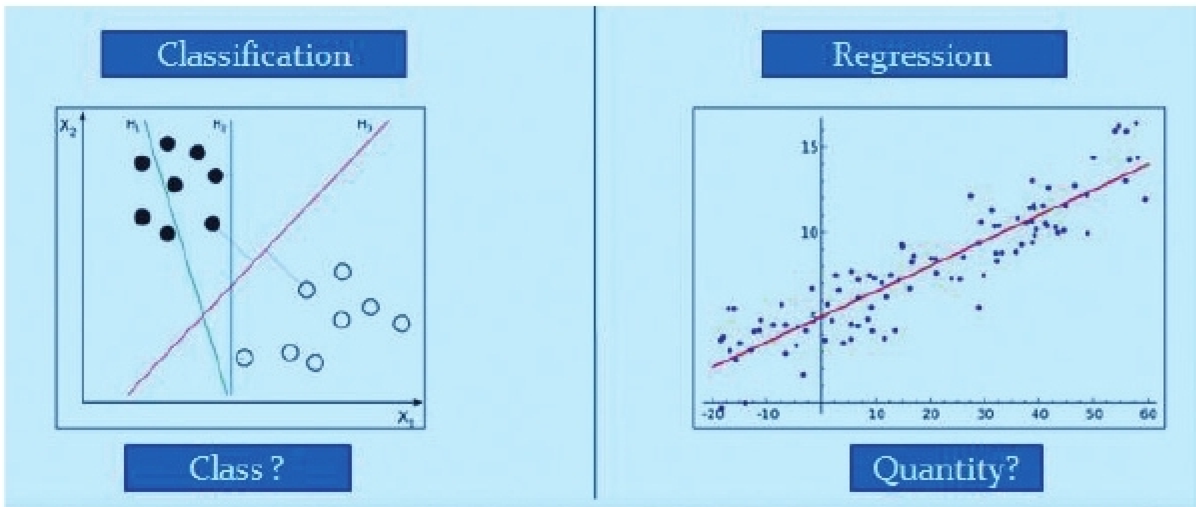

Figure 2b: Comparative view of supervised techniques

```
# Checking for null values in the dataset
dataset.isnull().values.any()
Out[5]:
True
In [6]:
## Checking which columns contain null values
print(dataset['Rank'].isnull().values.any())
print(dataset['Name'].isnull().values.any())
print(dataset['Platform'].isnull().values.any())
print(dataset['Year'].isnull().values.any())
print(dataset['Genre'].isnull().values.any())
print(dataset['Publisher'].isnull().values.any())
print(dataset['NA_Sales'].isnull().values.any())
print(dataset['EU_Sales'].isnull().values.any())
print(dataset['JP_Sales'].isnull().values.any())
print(dataset['Other_Sales'].isnull().values.any())
print(dataset['Global_Sales'].isnull().values.any())
```

Figure 3: Steps for data cleansing with pandas functions